

Understanding Classification

Classification is a type of supervised learning where the goal is to predict the **categorical label** (or "class") of a new observation based on past data. Unlike regression, which predicts continuous numbers, classification sorts things into buckets.

1. Decision Tree

A **Decision Tree** is a flowchart-like structure where each internal node represents a "test" on an attribute, each branch represents the outcome of the test, and each leaf node represents a class label. It splits the data into smaller and smaller subsets until it reaches a conclusion.

- **Example:** Imagine a bank deciding whether to approve a loan.
 - **Node 1:** Is the credit score > 700? (Yes / No)
 - **Node 2 (If Yes):** Is the income > \$50k? (Yes / No)
 - **Leaf:** If both are Yes, **Approve**. If any are No, **Reject**.

2. Random Forest

A **Random Forest** is an "ensemble" method. Instead of relying on one decision tree (which might be biased or too specific), it builds many different decision trees and lets them "vote." The class that gets the most votes across all trees becomes the final prediction.

- **Example:** You want to pick a movie to watch.
 - Instead of asking one friend (one Decision Tree), you ask ten friends (a Random Forest).
 - Seven friends suggest a Sci-Fi movie, while three suggest a Comedy.
 - You choose the **Sci-Fi** movie because it had the most votes.

3. Naive Bayes

Naive Bayes is based on Bayes' Theorem of probability. It is called "naive" because it assumes that every feature is completely independent of the others. It calculates the probability of a label given the input features and chooses the one with the highest probability.

- **Example:** Email Spam Filtering.
 - The algorithm looks for words like "Winner," "Free," and "Cash."
 - Even though these words often appear together, Naive Bayes treats them as independent signals.
 - If the probability of "Spam" given these words is higher than "Not Spam," the email is moved to the **Spam** folder.

4. K-Nearest Neighbors (KNN)

KNN works on the principle of proximity. It assumes that similar things exist in close proximity. To classify a new data point, it looks at the "\$K\$" closest points (neighbors) in the dataset and assigns the class that is most common among them.

- **Example:** Categorizing a new fruit.
 - You have a bowl of oranges (small, orange, rough) and apples (medium, red, smooth).
 - You find a new fruit that is small and orange.
 - If you look at the 3 nearest neighbors ($K=3$) and all 3 are oranges, the new fruit is classified as an **Orange**.